

Mission Control Button Hardening Report

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Objective

Begin the button-by-button hardening pass for Mission Control so visible actions are not fake, stale, or misleading.

Rule applied:

- A button may be live only when it maps to a real route.
- A read-only button may only call a read-only route.
- A protected action must be Bridge Session-gated or owner-approval-gated.
- A missing backend, missing credential, or unavailable integration must render as blocked/disabled with an exact blocker.
- No fake Done, fake live status, fake upload, fake send, or fake execution state is allowed.

Scope Completed In This Slice

This slice focused on the newly updated Gateway / Agent Hub / SpaceAgent Browser Automation surface and the central Mission Control button contract registry.

Changes Made

File	Change
`src/app/gateway/status/page.tsx`	Added a real read-only Gateway Status page so the Agent Hub header button no longer points at a loose route target.
`src/lib/button-contracts-route.test.ts`	Added a route-contract test that fails if any LIVE or READ_ONLY button contract points at a missing API route.

Button Contract Status

The existing `GET /api/bridge/button-contracts` contract already classifies Mission Control actions into allowed states:

- `LIVE`
- `READ\_ONLY`
- `BACKEND\_REQUIRED`
- `CREDENTIAL\_REQUIRED`
- `OWNER\_APPROVAL\_REQUIRED`
- `DISABLED`

The new test verifies:

- all `LIVE` and `READ\_ONLY` API endpoints physically exist in the Next.js route tree;
- blocked, credential-required, owner-approval-required, or disabled actions are not executable;
- `fake\_success\_allowed` is false for every contract.

Concrete Issue Found

Issue	Fix
The Agent Hub header had a `Gateway status` button targeting `/gateway/status`, but no dedicated page existed at that route.	Added `/gateway/status` as a real read-only production status page.

Gateway Status Page Behavior

The new `/gateway/status` page shows:

- Gateway status summary.
- Agent Zero status and blockers.
- Hermes status and blockers.
- Bridge / MCP status.
- OpenClaw+ / Build-Wiki / Farmer status.
- Button hardening rule.
- Safety confirmation.

It does not trigger writes, uploads, sends, Zapier, HeyGen, SMB, farmer execution, or connector execution.

Local Validation

```
| Check | Result |
|---|---|
| Direct TypeScript check with local binary | PASS |
| Focused Vitest suite | PASS, 4 files / 10 tests |
| Production build with Next.js | PASS |
| `.env` changes | none |
| Secrets printed | no |
| Raw auth files printed | no |
| Fake buttons added | no |
```

Focused tests run:

- `src/lib/button-contracts-route.test.ts`
- `src/lib/gateway-agent-hub.test.ts`
- `src/lib/space-agent-browser-automation.test.ts`
- `src/app/gateway-route-alias.test.ts`

Production Deployment

The production host was one commit behind the pushed branch. That is why the latest button hardening and `/gateway/status` page were not visible in Mission Control before this deployment.

```
| Item | Result |
|---|---|
| Production branch before deployment | `7e79944` |
| Production branch after deployment | `7d021b6` |
| Deployment mode | fast-forward pull from `sosastrike/to-knowledge-mc` |
| Production validation | PASS |
| Production restart | PASS |
| Old PID | `3205313` |
| Old start timestamp | `Fri May 8 07:46:23 2026` |
| New PID | `3228883` |
| New start timestamp | `Fri May 8 08:11:30 2026` |
| Standalone app bind | `127.0.0.1:3337` |
| Public local service exposure added | no |
| Playwright MCP bind | `127.0.0.1:8931` |
```

Production validation on the live host:

```
| Check | Result |
|---|---|
| `git diff --check` | PASS |
| `pnpm run typecheck` | PASS |
| `pnpm test` | PASS, 135 files / 1243 tests |
| `pnpm run build` | PASS |
```

| `GET /login` after restart | PASS |

Production Route Smoke

Authenticated API smoke used the existing production API-key auth path. No key value was printed.

Route	Authenticated result	Unauthenticated result
--- ---	--- ---	---
`GET /api/gateway/status`	200	401
`GET /api/gateway/registry`	200	not rerun unauth in this slice
`GET /api/gateway/agent-hub/status`	200	401
`GET /api/bridge/button-contracts`	200	401
`GET /api/bridge/agent-zero/status`	200	not rerun unauth in this slice
`GET /api/bridge/hermes/status`	200	not rerun unauth in this slice
`GET /api/gateway/nodes/pi`	200	not rerun unauth in this slice
`GET /api/bridge/pi/status`	200	401
`GET /api/bridge/playwright-mcp/status`	200	not rerun unauth in this slice
`GET /api/gateway/nodes/playwright-mcp`	200	not rerun unauth in this slice
`GET /api/gateway/space-agent/browser/status`	200	not rerun unauth in this slice
`GET /api/bridge/paperclip/status`	200	401

Page smoke used a temporary internal admin session created for HTTP verification and deleted after the check. This proves the deployed UI renders, but it is not claimed as owner-authenticated browser visual proof.

Page	Result	Proof
--- ---	--- ---	---
`GET /gateway/status`	200	renders Gateway Status and Button Contracts
`GET /gateway/agent-hub`	200	renders Agent Zero, Hermes, Pi, SpaceAgent, Paperclip, OpenClaw+, Playwright MCP, Firecrawl, and YouTube
unauthenticated `GET /gateway/status`	307	redirects instead of exposing the protected page

Action smoke:

Action	Result
--- ---	---
`GET /api/bridge/button-contracts` validation	PASS, 0 missing live routes and 0 invalid states
`POST /api/bridge/agent-zero/test-chat`	PASS, 200 with `agent_zero_called` present
`POST /api/bridge/playwright-mcp/smoke`	PASS, 200 with `ok:true`; local-only endpoint confirmed
`POST /api/bridge/hermes/test-chat`	BLOCKED, 503 with blocker `hermes_safe_live_chat_adapter_not_configured`

Additional Live Button Probe Finding

After the first production smoke, I ran the broader live button-contract probe. It found one real safety gap:

Finding	Resolution
--- ---	---
`POST /api/bridge/brain-sync/build-wiki/run-now`	could create an owner-channel approval request with operator authentication only. It did not start `opencloud-docs-farmer.service`, but it was still too permissive for the current hard rule that Build-Wiki/Farmer actions require an active scoped Bridge Session. Patched the route so it now returns `423` with blocker `active_bridge_session_require`

d_for_buildwiki_run_now` unless an active Agent Zero Bridge Session includes `buildwiki.run_now`. Added a regression test proving the route does not call the Telegram approval upstream when the Bridge Session is missing. |

Additional local validation for this patch:

```
| Check | Result |
|---|---|
| `src/app/api/bridge/brain-sync/build-wiki/run-now/route.test.ts` | PASS |
| `src/lib/button-contracts-route.test.ts` | PASS |
| `src/lib/gateway-agent-hub.test.ts` | PASS |
| `node scripts/check-button-contract-routes.mjs` | PASS, 53 API endpoints checked, 0 missing route files |
| `tsc --noEmit` | PASS |
| `next build --webpack` | PASS |
```

Production validation after deploying this patch:

```
| Check | Result |
|---|---|
| Production HEAD | `585da50` |
| Production restart after patch | PASS, PID `3248542`, start timestamp `Fri May 8 08:24:02 2026` |
| `node scripts/check-button-contract-live-status.mjs http://127.0.0.1:3337` | PASS, 52 live endpoints checked, 1 allowed skipped |
| `POST /api/bridge/brain-sync/build-wiki/run-now` without Bridge Session | PASS, 423 |
| Build-Wiki blocker | `active_bridge_session_required_for_buildwiki_run_now` |
| Approval created without Bridge Session | no |
| `GET /login` | 200 |
```

Remaining Honest Blockers

```
| Blocker | Status |
|---|---|
| Owner-authenticated browser visual proof | still requires the owner's actual browser/session; temporary admin smoke is not a substitute |
| Hermes live adapter | still blocked until `hermes_called:true` is proven |
| Firecrawl | still blocked until credential/backend is configured |
| YouTube Research | still limited until transcript connector is proven |
| Paperclip owner session bridge | still partial/degraded until owner login/session flow is proven |
```

No-Secrets Confirmation

- No tokens printed.
- No API keys printed.
- No auth files printed.
- No `.env` changes made.
- No SMB mount.
- No Fork 2.
- No Zapier writes.
- No HeyGen generation.
- No external farmers.
- No public local service exposure added.

Updated Decision

Gateway / Agent Hub button hardening: **GO for the deployed button-contract slice**.

Reason: the designer handoff UI is built and live on production, `/gateway/status`

s` now exists, all live/read-only button contracts have route proof, unauthenticated protected API routes reject access, and the safe Playwright MCP smoke passes.

Overall Gateway / Agent Hub remains ****PARTIAL GO**** until owner-authenticated visual proof is completed in the owner's actual browser.

Exact Next Step

Use the owner's authenticated browser session to visually confirm the production Agent Hub cards and action states. No code blocker remains for this slice.